

Project Proposal:

Enterprise E-Commerce Intelligence & Predictive Analytics

1. Project Description

This project leverages a **validated, proprietary dataset from a real-world global e-commerce corporation**, containing over 13,000 granular transaction records (including SKUs, PPC spend, and FBA fees). The initiative focuses on bridging the gap between massive data generation and executive decision-making.

By moving beyond static reporting, the project implements a **Modern Data Stack**, transitioning raw transactional data into a structured **Data Warehouse**. This foundation enables the application of **Machine Learning** to automate demand forecasting and optimize advertising ROI, providing a comprehensive roadmap for sustainable brand growth.

2. Team Composition

- **Team Leader & Strategic Lead:** Duaa Abd-Elati Abd-Elazeem – Project governance and stakeholder alignment.
- **Data Architect & Engineer:** Ahmed, Osama – Designing the **ETL (Extract, Transform, Load)** pipeline and Data Warehouse schema.
- **ML & Data Scientist:** Maya, Osama – Lead for predictive modeling, clustering, and statistical validation.
- **BI & Analytics Engineer:** Duaa, Hazem, Ahmed, Mai – Developing high-fidelity Power BI dashboards and semantic layers.

3. Integrated Project Objectives

The project is governed by a multi-layered analytical strategy, moving from foundational reporting to advanced predictive intelligence:

3.1 Foundational Analytics & Visualization

- **Trend & Performance Analysis:** Identify peak sales periods and evaluate individual product/category profitability using real-world transactional data.

- **Marketing Efficiency (PPC & Organic):** Calculate ROI for advertising campaigns and analyze the complex relationship between impressions, clicks, and actual conversion.
- **Strategic Visualization:** Develop a high-fidelity, interactive **Power BI KPI Dashboard** to support real-time executive decision-making and cross-departmental alignment.

3.2 Advanced Engineering & Machine Learning

- **Unified Data Architecture (SSOT):** Build a centralized **Data Warehouse** to unify disparate sales, marketing, and inventory metrics into a "Single Source of Truth."
- **Predictive Revenue Modeling:** Deploy advanced Machine Learning algorithms (e.g., **Random Forest, Prophet**) to forecast future sales and inventory requirements with a target accuracy of $\geq 90\%$.
- **Automated Profitability Guardrails:** Implement logic to identify "Bleeding SKUs" in real-time—products where FBA fees and PPC costs are aggressively eroding net margins.
- **Attribution & "Halo Effect" Analysis:** Quantify the impact of paid advertising on long-term organic search ranking and brand equity.

4. Strategic Intelligence & Actionable Insights

This phase transforms 13,000+ real-world corporate records into a multidimensional analysis. We categorize our insights into four high-impact pillars:

4.1 Product Intelligence & Portfolio Optimization

- The "Profitability vs. Volume" Matrix: Identifying high-volume "Hero" products vs. high-margin "Niche" products.
- Bleeding SKU Detection: Real-time identification of products where FBA Fees + PPC Cost > Gross Margin, allowing for immediate price or ad-spend adjustments.
- Market Basket Analysis: Utilizing Association Rule Learning (Apriori Algorithm) to identify which supplements (e.g., Magnesium + Zinc) are frequently bought together to inform bundling strategies.

4.2 Machine Learning & Predictive Analytics

- Demand Forecasting: Implementing Prophet or LSTM models to predict unit sales for the next 90 days, minimizing capital tied up in slow-moving inventory.

- **Anomaly Detection:** Automated alerts for sudden spikes in Refund % or drops in Unit Session %, signaling potential listing issues or competitor attacks.
- **Price Elasticity Modeling:** Calculating how a 5% change in price impacts total Net ROI.

4.3 Advertising (PPC) & "Halo Effect" Analysis

- **Total ACOS (TACoS) Benchmarking:** Measuring the true health of the brand by comparing total ad spend against *total* revenue (Organic + PPC).
- **The Halo Effect Attribution:** Statistical analysis to determine if a **€1.00** increase in PPC spend results in a measurable lift in **Organic Units** over a 30-day trailing period.
- **Efficiency Frontier:** Identifying the "Sweet Spot" for **Impressions vs. Clicks** to maximize conversion without overpaying for low-intent traffic.

4.4 Financial Risk & Operational Efficiency

- **Cost-Driver Decomposition:** Breaking down the "Net Margin" to see if losses are driven by logistics (FBA), marketing (PPC), or product quality (Refunds).
- **Marketplace Variance:** Comparative analysis of performance across different marketplaces (UK, DE, FR) to identify regional growth opportunities.

4.4 Profitability & Risk Management

- **Cost Drivers Analysis:** What are the primary factors (COGS, FBA Fees, or Promo Amounts) affecting profitability margins per product?
- **Price Elasticity & Sensitivity:** How does changing the product price impact total units sold and net profit?
- **Platform Dependency Risk:** To what extent is our profit dependent on a single marketplace or platform's fees.

5. Modern Tech Stack & Data Architecture

The project utilizes a scalable architecture to handle enterprise-level data processing:

- **SQL:** Extracting, filtering, and aggregating data from relational databases.
- **Excel:** Initial data auditing, cleaning, and basic calculations.
- **Python (Pandas, Matplotlib, Statsmodels):** Advanced statistical analysis and Time Series forecasting.
- **Power BI:** Designing interactive dashboards for visual storytelling.

6. Milestones & Deadlines

Milestone	Key Deliverables	Deadline
Phase 1: Data Prep	Cleaned dataset, Handled missing values/duplicates	Week 1
Phase 2: Exploratory Analysis	Calculation of core KPIs and descriptive statistics	Week 2
Phase 3: Advanced Analytics	Time Series Forecasting model and Correlation analysis	Week 3
Phase 4: Visualization	Final Power BI Dashboard development	Week 4
Phase 5: Documentation	Final Report and Presentation deck	Week 5

7. Key Performance Indicators (KPIs)

7.1 Business & Functional KPIs

- **Net Profit Margin:** $\frac{\text{Total Net Profit}}{\text{Total Revenue}}$ (Target: > 15%).
- **TACoS (Total ACOS):** $\frac{\sum \text{PPC Cost}}{\sum \text{Total Sales}}$ (Target: < 10% for mature brands).
- **Inventory Velocity:** Average units sold per day to optimize reorder points.

7.2 Project & Technical KPIs

- **Model Accuracy:** Mean Absolute Percentage Error (MAPE) $< 10\%$.
- **Data Integrity:** 100% reconciliation between raw source data and Warehouse tables.